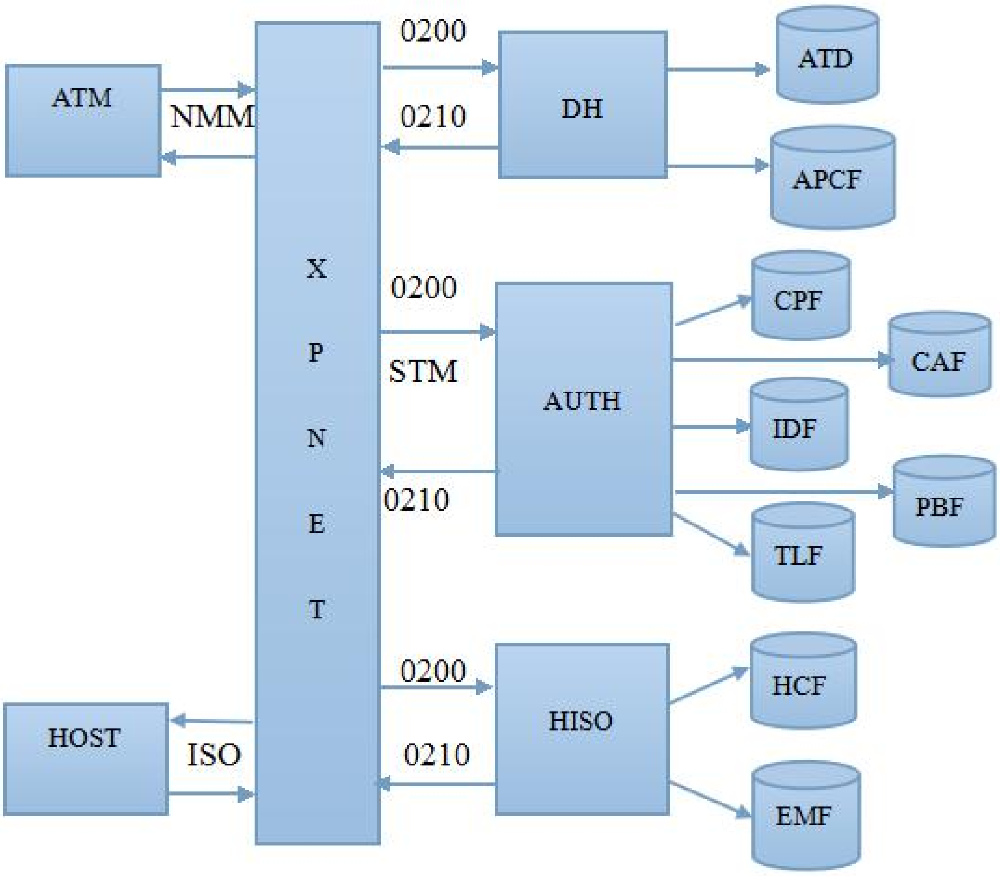
ATM On-Us Flow (Detail)



- **What is XPNET?**

Answer:

- The name **XPNET** is likely derived from its core functionality in **transaction processing and network management** within BASE24 systems. Though there is no officially documented full form, the name can be broken down as follows:
- **XP:** Possibly refers to "**Exchange Processing**" or "**Extended Processing**", indicating its role in handling financial transactions and routing.
- **NET:** Stands for "**Network**", as XPNET is primarily responsible for managing communication between ATMs, POS terminals, and host banking systems.
- **XPNET** is a network management and communication framework used in **BASE24** systems to handle transactions between ATMs, POS terminals, and banking hosts.
- It is responsible for **routing, switching, and monitoring financial transactions** in a secure and efficient manner.

Key Functions of XPNET:

- **Transaction Routing & Switching:**
  - Routes transaction messages (e.g., cash withdrawals, balance inquiries) between ATMs, POS terminals, and banking networks.
  - Supports multiple communication protocols, including **TCP/IP, X.25**, and **SNA**.

- **Network Management:**
  - Monitors and manages communication links between devices and host systems.

- Ensures high availability and fault tolerance in transaction processing.
- **Protocol Conversion:**
  - Converts messages between different network protocols (e.g., ISO 8583 over TCP/IP).
  - Helps in seamless integration with different banking networks.
- **Security & Encryption:**
  - Ensures secure communication of transaction data.
  - Works with cryptographic modules for PIN encryption and secure data transmission.
- **Real-time Monitoring & Logging:**
  - Provides logging and tracking of transactions for auditing and troubleshooting.
  - Generates alerts for failed or suspicious transactions.

How XPNET Works in BASE24?

- **ATMs & POS terminals** send transaction requests (like withdrawals or balance checks).
- **XPNET receives the request**, routes it to the appropriate processing system (BASE24 host or interbank network).
- **BASE24 processes the request** and sends a response back through XPNET.
- **XPNET ensures message delivery** and maintains session integrity using protocols like TCP/IP.

- **What is TCP/IP ?**

Answer:

- **TCP/IP (Transmission Control Protocol/Internet Protocol)**
  - TCP/IP is a set of networking protocols that allow computers to communicate over the internet and other networks. It serves as the foundation of modern networking and defines how data is sent, addressed, transmitted, routed, and received.
  - **Key Components of TCP/IP**
    - **TCP (Transmission Control Protocol)**
      - Ensures reliable, ordered, and error-checked delivery of data between applications.
      - Breaks data into packets, ensures they are sent and received in the correct order.
      - Retransmits lost packets if needed.
    - **IP (Internet Protocol)**
      - Handles addressing and routing of packets.
      - Ensures packets reach the correct destination based on IP addresses.
  - **Layers of TCP/IP Model**
    - TCP/IP is divided into four layers:
      - **Application Layer**
        - Handles communication between applications (e.g., HTTP, FTP, SMTP).
      - **Transport Layer**
        - Provides reliable (TCP) or faster but unreliable (UDP) communication.
      - **Internet Layer**

- Manages addressing and routing using IP.
- **Network Access Layer**
  - Deals with physical network connections (Ethernet, Wi-Fi).

- **Request:**
- Firstly a card is inserted in ATM and transaction is initiated
- **Que. What is Solicited and Unsolicited ?**
  - Ans.
    - Solicited : Idle State/ Ready State
    - Unsolicited : Not ready to take next transaction
    - ATM → NMM → DH
    - ↓
    - While cardholder performing transaction then ATM goes in unsolicited state
    - When any external event occurs or there is cash out error, general printer out error, power out then also ATM is in Unsolicited State.
  - NMM:

NMM	Unsolicited		Solicited
	Request	Reply	Status
Message Class	1	1	2
Message Subclass	1	4	2

- **Que. Why Unsolicited comes from ATM ?**

- ATM → DH
  - **Que. How request comes from ATM to DH?**
  - Ans.
  - ATM (Physical Device) → TCP/IP → Tandem contains NCS (Network Control Services) Utility → **Destination Attribute** of NCS contains which Logical Name/ Station Name (DH Process Name) of ATM and **RAddress (RAddr)** of NCS contains IP Address of ATM → DH
  - In ATM to DH request, following data gets transferred in NMM (Native Message Mode) Format :
    - Track 2
      - Contains:
      -
      - Primary Account Number (PAN), which is left justified (around 16-19 digits)
      - Field separator (=)
      - Country Code (if present; 3 Digits)
      - Expiration Date (YYMM/MMYY)
      - Service Code (if present; 3 digits)
      - **Que. What is Service code? (Which document should refer ? ATM Transaction Process)**
        - Ans: Service code values common in financial cards:
        - Card is of which type ? (Chip (999), Mag Strip (226/201) , Contactless)
      - Discretionary Data (Balance of available space) (PVV, CVV1)
      - **Que. What is CVV ? Difference between CVV1 and CVV2?**
        - CVV Stands for Card Verification Value

        - It is of 3 digits
        - **Que. What is iCVV (Integrated Chip CVV), dCVV (Dynamic CVV)?**
    - **Que. While Fallback transaction what will be the CVV value?**
    - **Que. What is PVV? What is Offline PVV and Online PVV?**
        - PVV Stands for PIN Verification Value
        - BASE-24 will not store PIN hence PVV is stored
        - PVV and PIN is matched using PIN Block
        - Key, Card Number, PIN → Algorithm → PVV is created → HSM matches this PVV with PVV stored in CAF
        - When both PVV Matches, then transaction is processed further
- Opcode
    - Opcode is converted into Trancode by DH
    - Trancode is searched by APCF to identify which type of transaction is done
    - **Que. What is Opcode and Trancode?**
- Amount (8 Byte length)
- PIN Block (4 digit plain PIN is converted into encrypted code of size 16 digits)
- **Que. What is PIN Block ? How it is encrypted and generated?**
- Last Transaction Details [Contains transaction status, dispense note count (Hopper Count), Last Transaction Sequence Number]
    - **Que. Why Last transaction details are used?**
        - Ans:
        - Last Transaction Details contains Transaction Sequence number of 4 bytes
        - That sequence number is matched with Last Transaction Sequence Number of ATD File.
            - If, that both sequence number matches then only transaction is processed further
            - else, transaction is declined (Error is there / Any Fraud)
        - This process is called as **Terminal Validation**
- Terminal ID (will be collected by Station Name)
- EMV
    - **Que. Which data is changed in EMV? (Why EMV is used ?)**
- **Que. How Request comes from ATM to DH?**
    - Ans: ATM (Physical Device) → TCP/IP Protocol → Tandem → NCS Utility (Contains ATM's Logical Name/Station Name) → Destination Attribute → DH Process Name → DH

- **Device Handler Processes**
    - DH will set the Timer (ATM → DH)
    - NMM is Converted into STM (The DH converts the NMM messages received from ATM to STM.)
    - **Que. What is NMM and STM?**
    - **Que. How NMM is converted to STM?**
        - Ans: STM's Stucture is :

Field 1	Field 2	Field 3	Field n		Tokens

- Each field contains data that to be sent to DH like Track 2, Trancode (6 bytes), Last Transaction Details, Amount, PIN Block, etc.
- Tokens Contains Extra Information
- **Que. What type of extra information is there into Tokens?**

- Terminal ID will be collected by Station Name
- DH process name will be extracted form the destination attribute which is available in NCPCOM Utility (NCS) (Which is on Tandem)
- DH will set the Timer (2 Timers are active, ATM → DH and DH → till Response Comes)
- **ATD (ATM Terminal Data File)**
  - Used for terminal validation (It checks whether the transaction received from particular terminal is valid terminal or not. by using Last Transaction Sequence Number)
  - Primary Key: Terminal ID
  - **Que. If Terminal ID is not available in ATD, then how it will decline?**
  - 2 Main Fields : DH Process name, Auth Process Name
  - It also contains hardware type, location, state, owner name on page no 1
  - It also contains Acquirer Transaction Profile (Which is a primary key for APCF) on page 2
  - It also contains Hoppers Count (Which is only used in response) on page no 4 and 5.
  - **Que. What does Hopper 0 Means? (Physical cash is present but hopper count is 0 in ATD)**
  - Page 6 shows terminal activity information.
  - It also contains last transaction details
  - Trancode is matched with APCF
    - If both matches, then transaction is proceed further
    - else, decline the transaction
  - ATD has 2 files : D1 (Dynamic) and S1 (Static)
  - **Que. What is difference between D1 and S1**
- **APCF (Acquirer Processing Code File)**
  - By using Acquirer transaction profile name DH reads the APCF file.
  - Contains configuration of transactions that an acquirer can do
  - Acquirers are ATM/POS/Interchange
  - Primary Key: Acquirer Transaction Profile
  - APCF file maps the trancode.
  - On page 2 it checks whether particular transaction is allowed for particular terminal or not.
  - Page 3 of APCF enables to load and unload multiple records simultaneously.
  - **Que. What is load and unload?**
- By using AUTH process name the DH sends the request message to particular AUTH via XPNET and sets the timer.

- **Authorization Processes**
  - **Que. What is Authentication (Prescreening) and Authorization?**
  - **Que. When and why Prescreening is done?**

- **Que. Prescreening is compulsory for everytime ?**
- DH sends the request message 0200 to AUTH.
- The AUTH first reads the CPF file
- **CPF (Card Prefix file)**
  - First 1 to 11 (Start reading from right) digits of card number is prefix (BIN)
  - This file contains BIN Level Information
  - Card Prefix is also known as BIN
  - Primary Key : BIN and PAN length
  - If BIN is present in CPF then it is an ON - US Transaction
    - then we will get Card FIID (Which is primary key of IDF)
  - If BIN is not present in CPF then it is an OFF - US Transaction then we will not get Card FIID
  - On page 1 we get FIID if it is on-us transaction and prefix routing.
  - **Que. What is Routing Table?**

DPC No.	Symbolic Name	Account Type	Prefix Routing	Auth Type	Auth Level
Primary Key for HCF	HISO Process Name	Savings Current etc	If we have to send a BIN to other Bank then it is used	Negative File UAF File (User Acceptance File) Starts from 0:Online 1:Neg File Only 2:CAF Only 3:CAF + PBF 4:Neg + UAF	Online Offline Onlie/Offline

  - **Que. Why 2 Routing Tables are maintained for ATM and PoS?**
  - Page 2 contains Pin verification information and Card verification information, check if host is online.
  - Page 3 verify the expiration date comparison, card validation period.
  - CPF also sets Card Activity Limits (Daily Limits) on page 4
    - It is BIN wise default limit which is common for all BIN
    - **It is set by Bank [CAF Activity Limit is set by Customers/Bank (Dynamic)]**
  - By using FIID, request proceeds to IDF
- **IDF (Institution Definition File)**
  - It contains the rules for bank (Financial Institution)
  - Primary Key: FIID
  - From IDF, we get path of CAF (1 File) and PBF (4 Files) on page 1.
  - On page 2, It performs pre-screening [Certain fields (Like Account Limit, CVV,

PVV, Expiry Date) are checked before sending the transaction to host]. In pre-screening only flags are there (Y/N) for checking that we have to check the details or not.

- In prescreening, it checks if host is online, maximum pin tries, expiration date verification, bad pin action.

- It gives DPC (Host Computer), Routing Table (Auth Level)
- **Que. What is Auth Level?**
- On screen 9 we get ATM routing table. By using prefix routing and account type match in the routing table the transaction is route to respective HISO by using DPC (Data Processing Center) number and HISO process name of respective match
- If the host is offline the BASE 24 approves the transaction hence it checks the CAF and PBF file
- By using FIID and PAN as primary key the AUTH reads the CAF file

- **CAF (Cardholder Authorization File)**
  - Contains card level data
  - CAF validates the card
  - Primary Keys: PAN, Member No., FIID
  - **Que. What is Member No.?**
    - Ans:
    - Card is almost unique for each cardholder.
    - But if card number is same then different member number is assigned.
    - It is of 3 digit
    - **Que. From where we can get Member Number?**
      - **Ans.**
      - **if , PAN Access Type = 1 then,**
        - **Member Number is taken from Track 2**
        - **Track 2 Sequence for Member Number is PAN → Member Number → Country Code → Expiry Date**
  - On page 1 of CAF it shows activity limits for individual card holder i.e card status, cash withdrawal limits.
  - **Que. What is Usage Limit?**
  - **Que. What is difference between Activity Limit and Usage Limit?**
  - Page 2 allows to set expiration date for individual card. It displays bad pin tries, date first used and last used date.
  - Page 3 & 4 shows multiple account for individual card holder.
  - Page 7 allows to set expiration date and shows card status for second card.
  - CAF Overrides CPF
  - CAF has 2 files : CAFD and CAF
  - **Que. What is difference between CAFD and CAF?**
  - **Que. If one card is linked with multiple savings account, then how CAF will know, from which account it should debit?**
  - **Que. During Fund Transfer, where does the From account & To account fields get populated.**
  - **Que. How account numbers are visible on ATM?**
- **PBF (Positive Balance File)**
  - Primary Key: FIID, Account Type, Account Number

- PBF file displays various balance and amount. (Account Details)
  - Page 1 keeps track of account balance. It shows last withdrawal date and amount, last deposit date and amount. Available balance and Account status.
- Balance Types :
  - Ledger Balance : Limit
  - Available Account Balance
  - Amount on Hold
- **Que. What is account status ?**

- **TLF (Transaction Log File)**
  - Primary Keys : 1) Card No.(PAN) & FIID 2) FIID & Terminal ID
  - It maintains the transaction logs in **Entry Sequence (FIFO)** manner
  - While response time the PBF, CPF, and CAF files are updated along with TLF. All transaction occurs at BASE 24 are logged in TLF file.
  - TLF logs the record in two format, terminal ID and PAN format. It displays trans code, transaction time and date, amount with respect to its particular terminal id or pan format.
- **Que. How to identify to which HISO to go ?**
  - **Ans.** With the help of routing table from IDF

- **Host Interface Processes (HISO)**
  - AUTH send the request message to respective HISO obtained in routing table from IDF file
  - **HCF (Host Configuration File)**
    - **Primary Key :** DPC Number, HISO Process Name
    - Host is a bank server (Computer)
    - Type of host, how many requests should sent, clock time is managed by HCF
    - From HCF file we get,
      - Page 1 displays timer limits for network management, store and forward (SAF), wait for traffic. It also displays processing flags, maximum SAF retry and data prefix characters.
      - **Que. What is SAF (Store and Forward)?**
      - **Que. What if SAF is not there ?**
      - Page 2 displays Station name associated with each DPC no. Station is a physical point in BASE 24 system that is designated to send and receive messages.
      - 2 stations are there.
      - Page 5 shows processing parameters and timer limits for ATM messages (i.e inbound and outbound limit, completion time and completion acknowledge).
      - **Que. What is inbound and outbound?**
    - HISO then reads the **EMF** file.
  - **EMF (External Message File)**
    - Primary Key: Interface Type, DPC Number, Process name, Prod, MSG - TYPE, In-Out-IND, Token Group, Full MSG MAC
    - It is based on Data Elements
    - Contains 3 elements :

        - M : Mandatory
        - C : Conditional
        - Blank/Null : Omit
    - Which data elements to be sent to host and to receive is decided by EMF. (If data elements is not received then transaction gets declined)
    - Page 1 allows to specify which data elements are to be included while sending and receiving external messages.
    - By using station name HISO sends the request to HOST and sets the timer.
    - The messages send by HISO are in ISO format

- **Response**
- HOST then sends the response to HISO which is in ISO format.
- **Que. How HOST knows to which HISO to be send the response?**
    - Ans.
    - There is a Suspense Buffer between HISO and HOST
    - It stores transaction address (Memory)
- **Que. What is Suspense Buffer?**
- **Que. How Host will know from which account to debit ? (Account Number's Data Element ?)**
- The HISO first checks the response is within the time if yes, then it sends the response to respective AUTH
- **Que. How HISO knows to which AUTH to be send the response ?**
    - Ans.:
    - STM contains Originator Process Name and Hold Originator Process Name
    - Auth Process Name is included in Originator Process Name
    - DH Process Name is included in Hold Originator Process Name
- **Que. When Auth ID Response (Data Element 38) is sent? (Request/Response)**
    - Ans. Response
- **Que. What is difference between Data Element 38 and 39?**
- **Que. Which Data Elements are sent in response?**
    - Ans. Transaction Amount Usage (4), Authorization Identification Response (38), Account Balance (54)
- AUTH then updates the TLF file. Along with TLF it updates the CAF and CPF limits and updates PBF.
- **Que. What will updated in PBF in response?**
- The AUTH sends the response message to DH. The DH checks if the timer if the response is within the time. DH then updates the ATD files hoppers count. It then sends the response to ATM.
- **Que. How hopper count gets updated in response?**
    - Ans: After user getting amount → DH gets solicited message by ATM → at that time Hopper count gets updated.
- ATM then checks the timer if response is within the time and completes the transaction successfully.
- ATM then sends the response message to DH which tells it has successfully completed the transaction and is ready for next transaction.
- **Que. What is RCPT?**
    - **Ans.**

- **RCPT** → Receipt Profile
- Receipt can get printed in two scenarios:
  - When requested before getting cash
  - When requested after getting cash
    - Though it is requested after getting cash and then we request for receipt, **it is considered in same transaction**
- **Que. From where we get Receipt Profile?**
  - **Ans. From ATD File (Which Page ?)**